

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

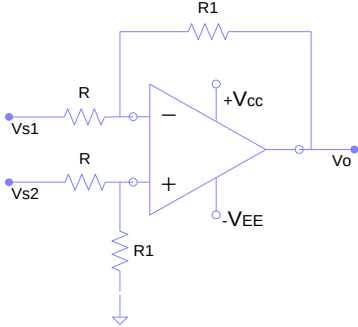
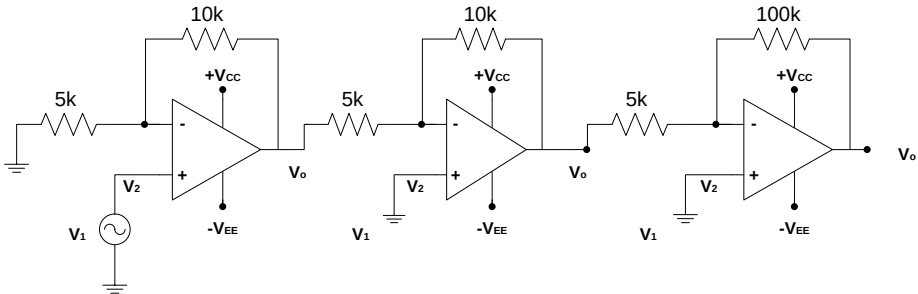
Department of Electrical Engineering

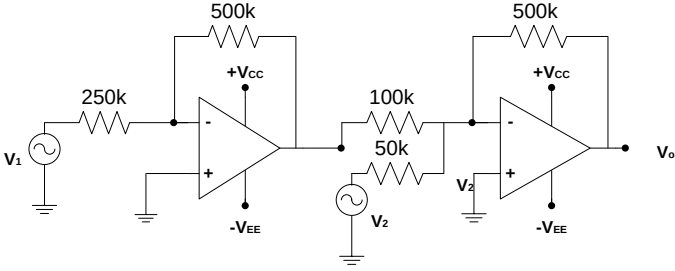
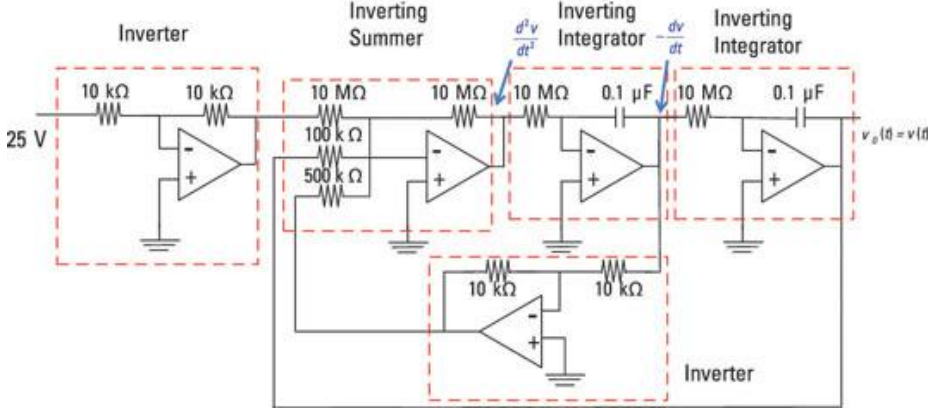
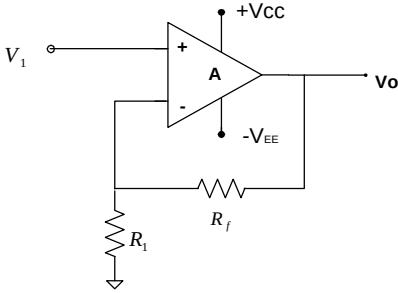
B. Tech. IV Semester

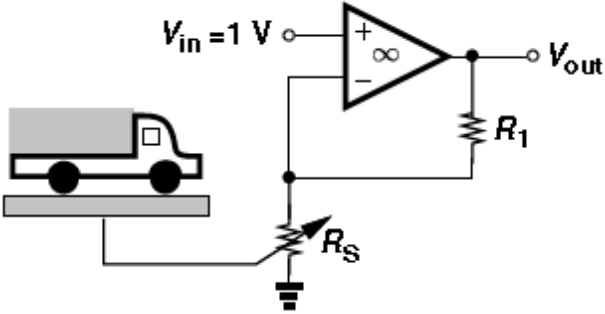
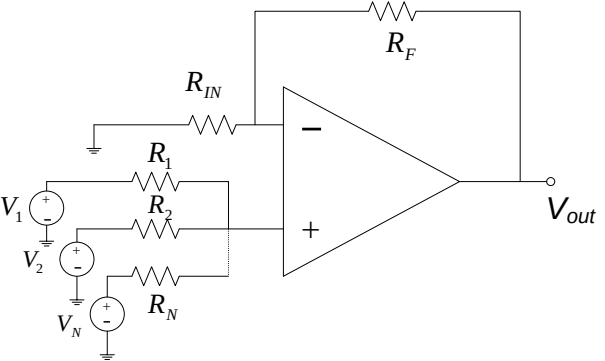
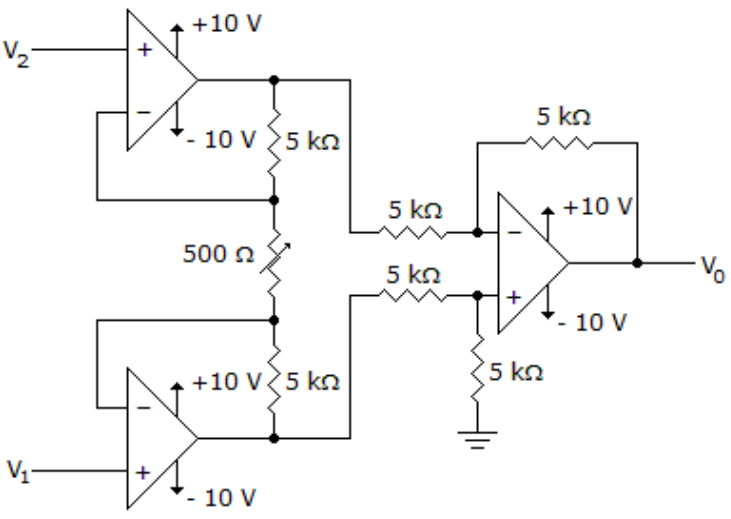
Sub: Electronics Devices & Circuits (EE-223)

Assignment – 2

Submission deadline: 31.03.2025

Q.1.	A Differential Amplifier has a CMRR of 90dB and a voltage gain of 20000. Estimate the common mode output for a common mode input of 10mV.
Q.2.	The OP Amp 741 has a slew rate of $2\text{V}/\mu\text{s}$. How long will the output take to change from 0 to 10 volts?
Q.3.	A 100 pf capacitor has a maximum charging current of $100\mu\text{A}$. What is the Slew rate? If the output of capacitor is applied to OP AMP 741, shall there be any distortion in the output?
Q.4.	Estimate the output voltage V_o for the circuit given in Fig. 1. Given $V_{s1} = 4\text{V}$, $V_{s2} = 2\text{V}$, $R_1=10\text{k}$ and $R=20\text{k}$.  Fig. 1
Q.5.	Explain the working of Schmitt Trigger circuit. A Schmitt trigger circuit is designed to have hysteresis window $V_H = 0.2\text{ volts}$, and feedback gain $\beta = 0.01$. Determine voltage V_R, UTP, and LTP.if reference voltage 'V_R' is half of the saturation voltage V_{Sat}.
Q.6.	Calculate the input voltage in Fig.2 below, if the final output voltage is 10 V. Assume all OP AMPs are ideal and biasing voltages are $\pm 12\text{V}$.  Fig. 2
Q.7.	Calculate the output voltage V_o for the circuit shown in Fig. 3. Given: $V_1 = -V_2 = 300\text{ mV}$

	 Fig.3
Q.8.	Discuss the operation of Schmitt trigger circuit in detail. In a Schmitt trigger circuit, $V_{Sat} = 8V$, $UTP = 4V$ and $LTP = 3V$. Determine β and V_R.
Q.9.	An OP AMP circuit used in analog computer is shown in Fig. 4. Prove that the circuit can be used to solve differential equation $\frac{d^2 v}{dt^2} = -20 \frac{dv}{dt} - 100 v + 25 V$  Fig. 4
Q.10.	Classify amplifiers based on input and output. Show that a circuit of Figure 5 is a case of series feedback amplifier. Calculate input resistance, output resistance, amplifier's gain and bandwidth for amplifier shown in Figure 4, if $A = 2 \times 10^5$ and $R_f = 20k$, and $R_i = 10k$  Fig. 5
Q.11.	The truck weighing-station in the Fig. 6 incorporates a sensor whose resistance varies as $R_s = R_0 + \alpha W$, where R_0 is a constant value, α is a proportionality constant & W the weight of the truck. If $V_{in} = 1 V$, determine the expression of gain, if gain of the system is defined as $\frac{dV_{out}}{dW}$.

	 Fig. 6
Q.12.	For the circuit in Fig. 7, If $R_1 = R_2 = \dots = R_N$, Prove that $V_{out} = \left(1 + \frac{R_f}{R_{in}}\right) \left[\frac{V_1}{N} + \frac{V_2}{N} + \dots + \frac{V_N}{N} \right]$ Where N is the number of the source connected  Fig. 7
Q.13.	The OP AMP circuit shown in Figure 8, is an example of Instrumentation Amplifier. Prove that the output of the circuit is adjustable with variable resistance of 500Ω, and for the given schematic of Figure 1, the output is 5.25 Volts.  Figure 8
Q.14.	An OP AMP circuit is shown in Figure 9. For $V_1 = -0.2$ Volts, and $V_2 = 0V$, find the output voltage.

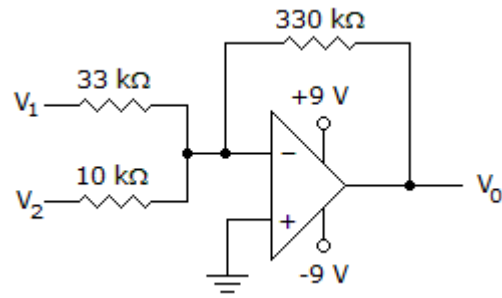


Figure 9

Q.15.

For the circuit shown in Fig. 10 below, show that V_0 is a linear function of δ if $\delta \ll 1$.

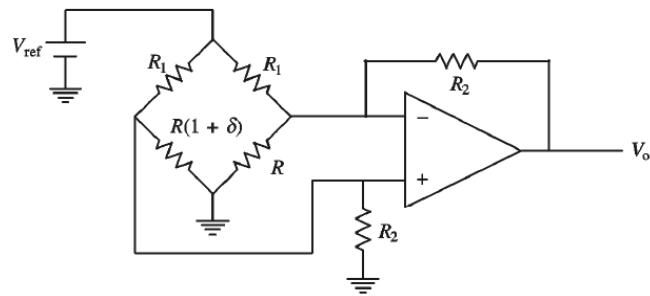


Fig. 10